

CSE 4402 Notes - Cryptography

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1 Syllabus / How to Class Works

- 6 homeworks. They are assigned whenever he feels like it.
- This class's content is different from MATH 4350

2 Introduction

2.1 The Communication Problem

- Typically, we have two characters who are trying to communicate a message (m) with each other, Alice (A) and Bob (B).
- However, there is Eve (E) who is trying to eavesdrop on the communication between A and B .
- We convert m to a crypt (c) via a key (k), send it to B , and then B decrypts it to get m .
- We assume that E can intercept the crypt (c) and try to build c s.t. she cannot decrypt it.
- **Def.** Kerchoff's Principle: Assume that E knows how our system works but doesn't know k .

2.2 Private Key Encryption Scheme

Let the following definitions be true:

- **Def.** m : message space. $m \in \mathbb{M}$
- **Def.** k : key space. $k \in \mathbb{K}$
- **Def.** c : cipher. $c \in \mathbb{C}$
- **Def.** Gen: Our key generation procedure (i.e. how we pick $k \in \mathbb{K}$) (assume uniformly at random)
- **Def.** Enc: Our encryption procedure. $\text{Enc}_k(m) \mapsto c$
- **Def.** Dec: decryption procedure. $\text{Dec}_k(c) \mapsto \{m, \perp\}$
 - Note that $\perp$ means 'null'.
- In this class, we display the probability of a given b as $P[b : a]$.
- **Def.** A procedure is 'valid' if $\forall m \in \mathbb{M}, P[k \leftarrow \text{Gen}; c \leftarrow \text{Enc}_k(m) : \text{Dec}_k(c) = m] = P_{k,c}[\text{Dec}_k(c) = m] = 1$

Ex. of a valid private key encryption scheme:

- Let m be 5-digit strings. Gen pick k at random.
- Hence, $P[\text{each key being selected}] = 1/10^5 = 10^{-5}$.
- $\text{Enc}_k(m)$: Return m concatenated with k (i.e. ' $m||k$ ')
- For example,

$$\text{Enc}_{10357}(BRIAN) = BRIAN10357$$

- Obviously, this is NOT very secure, so we should use a more secure scheme.

Ex. Caesar Shift:

- $\mathbb{M}$: finite strings of letters
- $\mathbb{K}$: $\{0, 1, \dots, 25\}$
- $\text{Enc}_k(m)$: Shift every letter in string forward by $\mathbb{K}$.
- $\text{Dec}_k(c)$: Shift every letter in string backward by $\mathbb{K}$.
- For example,

$$\text{Enc}_{k=3}(PIZZA) = SLCCD$$

- Obviously, this is NOT very secure, so we should use a more secure scheme.

Ex. Substitution Cipher:

- $\mathbb{M}$: finite strings of letters
- $\mathbb{K}$: permutations of $\{A, B, C, \dots, Z\}$
- $\text{Enc}_k(m)$: $\{A, B, C, \dots, Z\} \rightarrow \{A, B, C, \dots, Z\}$. Change every instance of letter x to $k(x)$.

- $\text{Dec}_k(c)$: change every instance of letter y to $k^{-1}(y)$.
- Breakable by frequency analysis (count the number of times each letter appears in the ciphertext and compare it to the frequency of each letter in the English language).
- Also vulnerable because certain letters are adjacent to other letters (ex. ‘Q’ is often associated with ‘U’).

2.3 Def. Shannon Secrecy

- Def. Shannon Secrecy: Given $\mathbb{M}$, $\mathbb{K}$, Gen, Enc, Dec, for any prior distribution, D on $\mathbb{M}$,

$$P_{k,m}[m = m_0 | \text{Enc}_k(m) = c] = P_m[m = m_0]$$